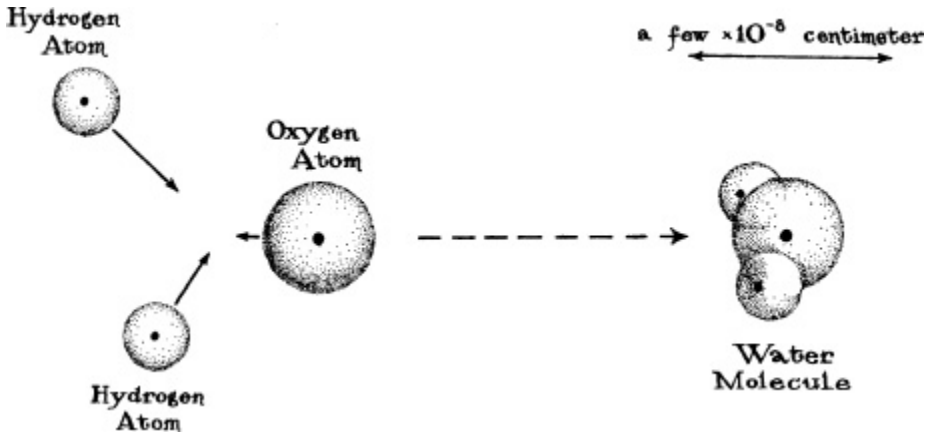
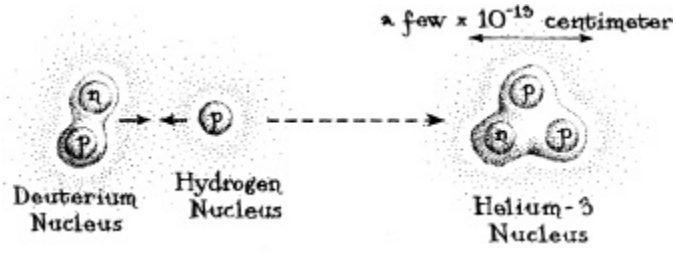


The following diagram shows an example of nuclear burning: the fusion of a deuterium (“heavy hydrogen”) nucleus and an ordinary hydrogen nucleus to form a helium-3 nucleus. This is one of the fusion reactions that is now known to power the Sun and other stars, and that powers hydrogen bombs (Chapter 6). The deuterium nucleus contains one neutron and one proton, bound together by the nuclear force; the hydrogen nucleus consists of a single proton; the helium-3 nucleus created by the fusion contains one neutron and two protons.



Capturing an atom onto a neutron core was much like dropping a rock onto a cement slab from a great height: Gravity pulls the rock down, accelerating it to high speed, and when it hits the slab, its huge kinetic energy (energy of motion) can shatter it into a thousand pieces. Similarly, gravity above a neutron core should accelerate infalling atoms to very high speeds, Landau reasoned. When such an atom plummets into the core, its shattering stop converts its huge kinetic energy (an amount equivalent to 10 percent of its mass) into heat. In this scenario, the ultimate source of the Sun’s heat is the intense gravity of its neutron core; and, as for Zwicky’s supernovae, the core’s gravity is 10 percent efficient at converting the mass of infalling atoms into heat.